SB3 · STATIC-MEMBRANE

Cook's membrane (skew panel, in-plane bending on a distorted mesh)

In-plane bending on a distorted mesh — custom membrane robustness check

PASS

error +0.184% · tol 3%

§2 TECHNICAL FEATURES TESTED

- **STRIX engine 4-node shell** — membrane with drilling rotations
- Production-default membrane wall, vertical XZ plane
- Skewed (trapezoidal) mesh — distortion test
- Combined in-plane bending + shear
- h-refinement convergence study

§3 PROBLEM DESCRIPTION

Cook's membrane is the standard distortion benchmark for in-plane elements: a tapered panel is **clamped on its left edge** A–D and loaded by a uniform shear traction on its right edge B–C, producing a combined bending–shear response. A uniform mesh of the trapezoid gives **skewed (non-rectangular) elements**, so how quickly the answer is recovered measures the element's robustness to mesh distortion. The monitored quantity is the vertical displacement at the loaded-edge midpoint Q, reported as the scale-independent normalized value $u_Q \cdot E \cdot t / P$ (the problem has no closed form; the reference is the plane-stress converged value). The panel is modelled at real structural scale and stands in the vertical XZ plane, as production walls do (panel x → global X, panel y → global Z); the plane problem and its answer are unchanged.

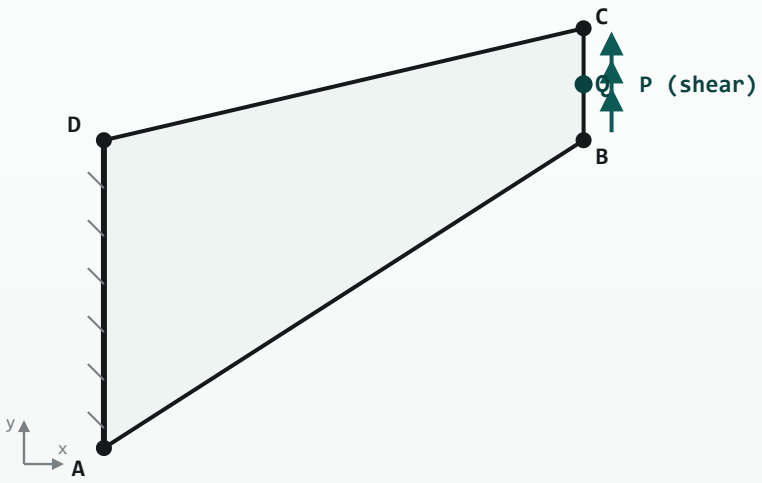


Figure SB3. Cook's skew panel — clamped edge A–D, uniform shear on B–C, monitored at the loaded-edge midpoint Q.

§4 GEOMETRY, PROPERTIES & LOADING

GEOMETRY		MATERIAL		LOADING & BC	
Corners (unit)	A0,0 B48,44 C48,60 D0,44	Model	Elastic · plane stress	Shear on B–C	1×10⁶ N (panel +y = global +Z)
Scale	100 × (mm)	E	30000 MPa	Clamp	u_x=u_y=0 (A–D) — panel axes
Probe Q	mid edge (48,52)	ν	0.333333	Out-of-plane	global u_Y, r_X, r_Z fixed; r_Y free
Thickness t	200 mm	Units	N · mm	Analysis	Linear static

§5 REFERENCE SOLUTION

Normalized tip displacement u_Q*E*t/P (mid loaded edge): u_y(Q)*E*t/P, Q = midpoint of loaded edge (plane stress, nu=1/3) = 23.91 norm.

*Cook's membrane, a standard finite-element distortion benchmark (R.D. Cook, 1974). No closed-form solution; the reference is the fine-mesh converged tip displacement at the loaded-edge midpoint Q, PLANE STRESS, E=1, nu=1/3, unit shear resultant: normalized u_Q*E*t/P = 23.91 (literature band 23.90-23.96). The plane-strain counterpart is 21.52 (Sevilla et al. / locking-free FCFV report 21.42 converged); STRIX walls are plane stress (membrane section), and 21.42/(1-nu^2) ~= 23.9 confirms the plane-stress mid-edge value.*

§6 RESULTS COMPARISON

RESPONSE QUANTITY	PROBE	STRIX	REFERENCE	DIFF	VERDICT
Normalized tip displacement u_Q*E*t/P (mid loaded edge) (norm.)	mesh 64×64	23.9541	23.91	+0.184%	PASS

§7 CONVERGENCE

MESH	U_Q*E*T/P (NORM.)	DIFF
2×2 cells	20.4913	−14.298%
4×4 cells	22.8958	−4.242%
8×8 cells	23.6462	−1.103%
16×16 cells	23.8663	−0.183%
32×32 cells	23.9325	+0.094%
64×64 cells	23.9541	+0.184%

On the distorted trapezoidal mesh the STRIX engine membrane response `rises monotonically` with refinement: −14.3% on the 2×2 mesh, −4.24% on 4×4 and −1.10% on 8×8, then within 0.2% of the plane-stress reference from 16×16 on, finishing at +0.18% on the finest 64×64 mesh — slightly above 23.91 and inside the 23.90–23.96 range of published converged values.

§8 CONCLUSION

PASS On Cook's distorted mesh the normalized mid-edge displacement $u_Q \cdot E \cdot t/P$ rises monotonically toward the plane-stress reference of 23.91 and the finest mesh lies well within tolerance. The STRIX engine wall shell is verified to be robust to mesh distortion in combined in-plane bending and shear.

§9 REFERENCES & ANALYSIS FILES

1. R.D. Cook (1974) — skew cantilever ("Cook's membrane"), the standard mesh-distortion benchmark. No closed-form solution.
2. Plane-stress reference (ν=1/3, unit shear): normalized $u_Q \cdot E \cdot t/P \approx 23.90–23.96$ at the loaded-edge midpoint. Plane-strain counterpart 21.52 (Sevilla et al. / FCFV report 21.42 converged); $21.42/(1-\nu^2) \approx 23.9$ fixes the plane-stress value.
3. STRIX headless harness: `generatePyForLoadCase → STRIX engine (static-single route) → node_disp.out`, production-default membrane wall.

EVIDENCE ARCHIVE `verif-evidence-strix0.134.1-win-x64.zip · SB3/` `sha256 (pending publish)`